AI Researcher Take-Home Assessment

As an AI Researcher, your role is to move beyond theory and help turn ideas into products by exploring new paradigms, evaluating technical trade-offs, and building robust autonomous systems.

This assignment evaluates your ability to design an agentic workflow, implement advanced evaluation methodologies, and compare the performance of frontier closed-source models against leading open-source models.

Part 1: Research & Model Evaluation

- **The Task:** Select one closed-source frontier model (from major AI providers such as OpenAI, Anthropic, or Google) and one leading open-source AI model.
- **The Methodology:** Implement a trajectory-based evaluation technique to assess how each model handles an agentic workflow. Instead of just looking at the final output, evaluate intermediate reasoning, tool-calling accuracy, and error recovery.
- **Emerging Techniques:** Identify and document novel techniques, architectures, or frameworks (such as graph-based state management, advanced multi-agent orchestration, process mining, or specialized skill integrations like Claude Skills) that could be utilized to build and optimize these proactive, self-healing systems.
- **The Deliverable:** A structured research report documenting the technical trade-offs, benchmarking results, experimentation methodologies used to compare the two models, and a review of the new techniques identified for building the system.

Part 2: Technical Proof-of-Concept (POC)

- **The Task:** Utilize an AI Agent framework such as LangGraph, LangChain, CrewAI, AutoGen, or a similar platform to build a lightweight, multi-agent workflow.
- **The Use Case:** The system must solve a specific business problem, such as autonomous carrier rerouting during a logistics disruption. It should ingest a telemetry alert (like a delay or failure), evaluate alternatives, and autonomously execute a reroute without human intervention.
- **The Deliverable:** A complete code repository using Python and relevant AI development tools. The repository must demonstrate practical AI-powered automation and workflow design.

Part 3: Product Strategy & Implementation Presentation

- **The Task:** Develop a presentation that translates your technical research findings into actionable product recommendations.

- **The Focus:** Connect your technical findings with real business and product opportunities, explaining whether the open-source model is mature enough to replace the closed-source model for this specific use case in a production environment.
- **The Deliverable:** A 15-minute presentation deck designed to explain complex AI concepts clearly to both technical and non-technical audiences.

Assignment Guidelines & Policies

- **Time Expectation:** We respect your time. Please limit your work on this assignment to 3–4 hours. We are evaluating clarity of reasoning over code quantity, and sound methodology over model complexity.
- **AI Tool Policy:** This is an open-book assessment. You are encouraged to use AI coding assistants (e.g., Copilot, ChatGPT, Cursor). However, you must include a short decision log in your README explaining how you used these tools, what you accepted/rejected, and the architectural decisions you made by hand.
- **Submission Format:** Please share your code in a well-organized Git repository (GitHub/GitLab). Ensure your README includes setup instructions so our team can easily replicate your environment and run your local pipeline.
- **Evaluation Rubric:** You will be assessed on problem framing, technical communication, the ability to justify architectural trade-offs, and your experimentation mindset.